

Solar Photovoltaic Entrepreneur

Qualification Pack: SGJ/Q0901 | NSQF Level: 4 | Total Duration: 30 Hours (1,800 Mins)

Module 1: Introduction to Solar PV Sector in India (3.0 Hours)

UNIT 1.1 Overview of Solar PV Technology (0.75 Hours / 45 Mins)

- 1.1.1 Introduction
- 1.1.2 Solar Photovoltaic (PV) System
- 1.1.3 Solar PV Technology
- 1.1.4 Overview of Solar PV technology, Rooftop and Ground Mounted Solar, Grid Interactive and Off- Grid Solar Sector in India

UNIT 1.2 Broader Business, Finance, Technology and Regulatory Landscape of Solar Energy in the Country (0.75 Hours / 45 Mins)

- 1.2.1 Market Research
- 1.2.2 Business Landscape
- 1.2.3 Finance Landscape
- 1.2.4 Technology Landscape
- 1.2.5 Regulatory Landscape
- 1.2.6 Benefits of Solar Energy Over Conventional Sources of Energy

UNIT 1.3 Overview of Grid Interactive and Off-Grid Solar Sector in India (0.50 Hours / 30 Mins)

- 1.3.1 Overview of Grid Interactive and Off- Grid Solar Sector in India
- 1.3.2 Types of off-grid solar PV power devices and their working principles

UNIT 1.4 Basics of Electrical Concepts (0.50 Hours / 30 Mins)

- 1.4.1 Current
- 1.4.2 Direct Current (DC) and Alternating Current (AC)

UNIT 1.5 Financial Institutions and Banks Involved in Financing Solar Power Projects (0.50 Hours / 30 Mins)

- 1.5.1 Financial Institutions and Banks Involved in Lending
- 1.5.2 Terms and Conditions Associated with Loans

Module 2: Set up new venture for Solar Photovoltaic System (3.0 Hours)

UNIT 2.1 Setting Up a New Venture (0.50 Hours / 30 Mins)

- 2.1.1 Entrepreneurship
- 2.1.2 Process of Setting Up a New Venture
- 2.1.3 Evolving opportunities across solar solutions, services and market ecosystem
- 2.1.4 Key Ingredients of a Business Plan
- 2.1.5 Components of Loan Application

UNIT 2.2 Workplace Etiquette (0.50 Hours / 30 Mins)

- 2.2.1 Forwarding Accurate Information
- 2.2.2 Assisting Others
- 2.2.3 Interact with Others without Any Bias
- 2.2.4 Communicating Effectively
- 2.2.5 Interacting Effectively with Clients & Colleagues

UNIT 2.3 Time Management, Leadership Skills and Resource Management (0.50 Hours / 30 Mins)

- 2.3.1 Time Management
- 2.3.2 Effective Time Management Techniques
- 2.3.3 Leadership Qualities That All Entrepreneurs Need
- 2.3.4 Resource Management

UNIT 2.4 Productivity Tools: MS Word, MS Excel, PowerPoint and Presentations (0.50 Hours / 30 Mins)

- 2.4.1 MS Word
- 2.4.2 MS Excel

2.4.3 MS PowerPoint

UNIT 2.5 Challenges and Risks for New Entrepreneurs (0.25 Hours / 15 Mins)

2.5.1 Challenges and Risk for New Entrepreneurs

2.5.2 Choose the Right Buyer

UNIT 2.6 Taxes and Compliances (0.25 Hours / 15 Mins)

2.6.1 Tax and Related Compliances

UNIT 2.7 Aspects of Successful Solar Business (0.25 Hours / 15 Mins)

2.7.1 Key Aspects to Successfully Run a Solar Business

UNIT 2.8 Respecting Gender and Disability Concerns (0.25 Hours / 15 Mins)

2.8.1 Respecting Diversity and Gender

2.8.2 Working in a Diverse Environment

2.8.3 Gender Sensitivity and PwD (Person with Disability)

Module 3: Business Model in Solar Photovoltaic (3.0 Hours)

UNIT 3.1 Indian Solar Industry Overview (0.75 Hours / 45 Mins)

3.1.1 Solar Industry Overview

3.1.2 Key Stakeholders

3.1.3 Stakeholder Roles

3.1.4 Market Drivers

UNIT 3.2 Solar Business Models (1.00 Hour / 60 Mins)

3.2.1 Solar Business Models

3.2.2 Comparison of Solar Business Models

3.2.3 CAPEX Model

3.2.4 OPEX / RESCO Model

3.2.5 Lease Model

3.2.6 BOOT Model

3.2.7 Hybrid Model

3.2.8 Community Solar and Shared Solar

UNIT 3.3 Financial Performance Indicators (0.75 Hours / 45 Mins)

3.3.1 Payback Period

3.3.2 Internal Rate of Return (IRR)

3.3.3 Net Savings

3.3.4 Practical Activity - Excel IRR() function and simple payback period

UNIT 3.4 Policy and Regulatory Framework (0.50 Hours / 30 Mins)

3.4.1 Role of MNRE

3.4.2 State Nodal Agencies

3.4.3 DISCOM Regulations - Net Metering; Grid Connectivity; Billing and Energy Accounting; Safety Compliance

Module 4: Site Feasibility Study for Rooftop Solar power plant (3.0 Hours)

UNIT 4.1 Site Assessment and System Sizing (1.75 Hours / 105 Mins)

4.1.1 Identifying the Optimum Rooftop Location

4.1.2 Factors Affecting Site Selection

4.1.3 Site-Level Pre-Requisites

4.1.4 Selection of Mounting Structure

4.1.5 Shading Assessment

4.1.6 Basic Site Mapping

4.1.7 Load Assessment and Load Profile

4.1.8 Estimating Solar Plant Capacity

UNIT 4.2 Solar Resource Assessment, Energy Yield and Feasibility Report (1.25 Hours / 75 Mins)

- 4.2.1 Battery Backup Assessment
- 4.2.2 Solar Resource Assessment
- 4.2.3 Energy Generation Estimation
- 4.2.4 Risk Assessment
- 4.2.5 Site Feasibility Study Report

Module 5: Site Feasibility Study for Ground Mount Solar power plant (3.0 Hours)

UNIT 5.1 Site Feasibility (0.75 Hours / 45 Mins)

- 5.1.1 Assess Irradiation and Other Geological Data
- 5.1.2 Albedo
- 5.1.3 Collecting data on local weather conditions

UNIT 5.2 Soil Testing (0.50 Hours / 30 Mins)

- 5.2.1 Tests to evaluate the soil at the site - Soil Composition; Soil Bearing Capacity; Soil Resistivity; Soil pH
- 5.2.2 Some Common Methods for Soil Testing
- 5.2.3 Note / reference video: "How to conduct SPT"

UNIT 5.3 Topography, Contour Mapping and Its Importance in Solar Plant Design (0.50 Hours / 30 Mins)

- 5.3.1 Importance of Topography in Solar Design
- 5.3.2 Contour Mapping and its Importance in Solar Design

UNIT 5.4 Electrical Feasibility Report for Ground Mount Solar Power Plant (0.75 Hours / 45 Mins)

- 5.4.1 Electrical Feasibility
- 5.4.2 Assessing Site Electrical Feasibility
- 5.4.3 Assessing Cable Root
- 5.4.4 Identifying Inverter Location
- 5.4.5 Identifying Transformer and Other HT equipment location

UNIT 5.5 Estimate Energy Generation from the Solar PV Power Plant (0.50 Hours / 30 Mins)

- 5.5.1 Estimate Energy Generation
- 5.5.2 Assess Grid Availability
- 5.5.3 Assess the Evacuation Voltage Level

Module 6: Installation of Solar Rooftop PV Plant (3.0 Hours)

UNIT 6.1 Civil, Mechanical and Electrical Installation (1.75 Hours / 105 Mins)

- 6.1.1 Civil, Mechanical and Electrical Installation
- 6.1.2 Table 6.1.1: Major Components Installed in a Rooftop Solar PV Plant
- 6.1.3 Table 6.1.2: Common Tools Used During Installation

UNIT 6.2 Installation Supervision, Testing and Commissioning (1.25 Hours / 75 Mins)

- 6.2.1 Installation Supervision, Testing and Commissioning

Module 7: Installation of Solar Ground Mounted PV Plant (3.0 Hours)

UNIT 7.1 Interpret and prepare Civil foundation, layout, mounting structure and allied civil work in ground mount solar plant (1.00 Hour / 60 Mins)

- 7.1.1 Project Development Plan
- 7.1.2 Assess Soil Characteristics
- 7.1.3 Type of Foundations
- 7.1.4 How to Prepare Structural Design Report
- 7.1.5 Preparing the Layout for Cable Trenching, Electrical Control Panel, Transformer, and VCB
- 7.1.6 Preparing Pile Foundation, Column Post and Other Civil Foundation
- 7.1.7 Assess Design of the Civil/ Structural Allied Works of the Solar PV Power Plant
- 7.1.8 Civil/ Structural Drawings for the Ground Mount Solar Power Plant
- 7.1.9 Discuss the Reinforcement Material for Civil Construction of Various Foundation

UNIT 7.2 Manage Ground Mounted Solar Civil Construction work (1.00 Hour / 60 Mins)

- 7.2.1 Project Development Plan for Ground-Mounted Solar Civil Work Construction
- 7.2.2 Construction of Compounds Wall/Fencing, Access of Internal Road, Drainage
- 7.2.3 Overall Project Constructability Approach for Cost Effective Implementation
- 7.2.4 Plan, Schedule, and Monitor All Material Deliveries to Avoid Project Delays
- 7.2.5 Supervise and Manage
- 7.2.6 Maximize Profit
- 7.2.7 Manage Project Billing Cycle
- 7.2.8 Analysing Bids
- 7.2.9 Negotiation Skills
- 7.2.10 Communicating with Customers and Team Members
- 7.2.11 Periodic Site Visits

UNIT 7.3 Interpret drawings of DC and AC side of electrical Component of Ground Mount Solar Power Plant (0.50 Hours / 30 Mins)

- 7.3.1 Interpretation of SLD and interconnections
- 7.3.2 Electrical line diagram of ACDB, LT Panel, earthing layout

UNIT 7.4 Manage Electrical installation in Ground Mount Solar Plant (0.50 Hours / 30 Mins)

- 7.4.1 Identify manufacturers and suppliers through various sources
- 7.4.2 How to manage vendor or supplier for material procurement
- 7.4.3 Know-how of all electrical tools utilised for system installation
- 7.4.4 Testing and commissioning of equipment in DC and AC side of the plant

Module 8: Operation and maintenance of Solar Rooftop PV Plant (3.0 Hours)

UNIT 8.1 Tools Required for PV System Maintenance (0.75 Hours / 45 Mins)

- 8.1.1 List of Tools for Maintenance Activities
- 8.1.2 Maintenance Schedule and Logbook
- 8.1.3 Important to Note

UNIT 8.2 Preventive Maintenance of PV System (1.25 Hours / 75 Mins)

- 8.2.1 What is a Maintenance Schedule?
- 8.2.2 PV Module Maintenance
- 8.2.3 Monitor Power Generation
- 8.2.4 Maintenance of Batteries
- 8.2.5 Maintenance of Inverter
- 8.2.6 Maintenance of Cables, Connectors and Switches

UNIT 8.3 Troubleshooting and Maintenance (1.00 Hour / 60 Mins)

- 8.3.1 Troubleshooting with Batteries
- 8.3.2 Troubleshooting with Charge Controllers
- 8.3.3 Troubleshooting with Inverters
- 8.3.4 Troubleshooting with Solar PV Arrays

Module 9: Operation and maintenance of Solar Ground Mounted PV Plant (3.0 Hours)

UNIT 9.1 Operation and Maintenance of Solar Ground Mounted PV Plants (1.75 Hours / 105 Mins)

- 9.1.1 List of Tools for Maintenance Activities
- 9.1.2 Maintenance Schedule and Logbook
- 9.1.3 Table 9.1 Types of Maintenance in Solar Ground Mounted PV Plants

UNIT 9.2 Inspection, Troubleshooting and Performance Optimization (1.25 Hours / 75 Mins)

- 9.2.1 Inspection, Troubleshooting and Performance Optimization

Module 10: Maintain Personal Health & Safety at Project Site (3.0 Hours)

UNIT 10.1 Workplace Safety (1.00 Hour / 60 Mins)

- 10.1.1 Potential Sources of Accidents

- 10.1.2 Safety Policies
- 10.1.3 General Safety Guidelines
- 10.1.4 Safety Measures and Precautions
- 10.1.5 Reporting Incidents
- 10.1.6 Precautions for Fire Hazards
- 10.1.7 Fire Drill
- 10.1.8 Using Fire Extinguisher

UNIT 10.2 Workplace Hazards (0.50 Hours / 30 Mins)

- 10.2.1 Workplace Hazards
- 10.2.2 Health and Sanitation Requirements at Workplace

UNIT 10.3 Standard Safety Practices (0.75 Hours / 45 Mins)

- 10.3.1 Personal Protection Practices
- 10.3.2 Standard Safety Operations at Worksite
- 10.3.3 Improvements in Health and Safety at Workplace

UNIT 10.4 Safety Actions (0.75 Hours / 45 Mins)

- 10.4.1 Administer First Aid
- 10.4.2 Report of Periodic Checks
- 10.4.3 Documented Safety Protocols at Workplace
- 10.4.4 Housekeeping